Proof by Mathematical Induction

Section 1: The Principle & Architecture of Induction

1 In a proof by mathematical induction that a proposition $P(n)$ is true for all integers $n \geq 1$, what are the essential logical components?

a. Base case: prove $P(1)$ is true; Inductive step: prove $P(k) \Rightarrow P(k+1)$; Conclusion by PMI

b. Verify $P(n)$ for $n = 1, 2, 3, 4, 5$ and deduce the pattern

c. Assume $P(n)$ is true for all n and differentiate both sides

d. Prove $P(k+1) \Rightarrow P(k)$ starting from an arbitrary large integer

2 If the base case of an induction proof is verified at $n = 4$, for which set of integers is $P(n)$ established (assuming the inductive step holds for all $k \geq 4$)?

a. $n \in \{4, 5, 6, \ldots\}$

b. All integers $n \in \mathbb{Z}$

c. $n \in \{1, 2, 3, 4\}$

d. $n \in \mathbb{R}$ with $n \geq 4$

3 In proving $\sum_{r=1}^{n} r = \frac{n(n+1)}{2}$, what is the algebraic expression for the sum up to $k+1$ using the induction hypothesis?

a. $\frac{k(k+1)}{2} + (k+1)$

b. $\frac{(k+1)(k+2)}{2} + k$

c. $\frac{k(k+1)}{2} \times (k+1)$

d. $\frac{k(k-1)}{2} + (k+1)$

Section 2: Summation and Series Proofs

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4 When proving by induction that $\sum_{r=1}^{n} r^2 = \frac{n(n+1)(2n+1)}{6}$, factoring $(k+1)$ from $\frac{k(k+1)(2k+1)}{6} + (k+1)^2$ yields:

a. $\frac{k+1}{6}[2k^2+7k+6] = \frac{(k+1)(k+2)(2k+3)}{6}$

b. $\frac{k+1}{6}[2k^2+5k+6]$

c. $\frac{(k+1)^2(2k+1)}{6}$

d. $\frac{k+1}{6}[k(2k+1)+1]$

5 For the proposition $\sum_{r=1}^{n} (2r-1) = n^2$, verify the inductive step by simplifying $k^2 + (2(k+1)-1)$.

a. $(k+1)^2$

b. k^2+2k

c. k^2+2k+3

d. $(k+2)^2$

6 Consider the geometric series $\sum_{r=1}^{n} 2^{r-1} = 2^n - 1$. What is the base case check for $n=1$?

a. $\text{LHS} = 2^0 = 1$, $\text{RHS} = 2^1 - 1 = 1$; $\text{LHS} = \text{RHS}$

b. $\text{LHS} = 2^1 = 2$, $\text{RHS} = 2^1 - 1 = 1$

c. $\text{LHS} = 0$, $\text{RHS} = 0$

d. $\text{LHS} = 1$, $\text{RHS} = 2$

Section 3: Divisibility and Inequality Proofs

7 To prove $5^n - 1$ is divisible by 4 for $n \geq 1$, express $5^{k+1} - 1$ in terms of $(5^k - 1)$.

a. $5(5^k-1)+4$

b. $5(5^k-1)-4$

c. $4(5^k-1)+5$

d. $(5^k-1)+5$

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8 In proving $2^n > n^2$ for $n \geq 5$, assuming $2^k > k^2$, which inequality must be justified to show $2^{k+1} > (k+1)^2$?

a. $2k^2 \geq (k+1)^2 \Leftrightarrow k^2 - 2k - 1 \geq 0$, which holds for $k \geq 5$

b. $k^2 > 2k + 1$ for all $k \in \mathbb{R}$

c. $2^k > 2k$

d. $(k+1)^2 > k^2 + 1$

9 To prove $3^{2n} - 1$ is divisible by 8, evaluate the base case $n = 1$.

a. $3^2 - 1 = 8$, which is divisible by 8

b. $3^1 - 1 = 2$

c. $3^0 - 1 = 0$

d. $3^2 - 1 = 9$

Section 4: Calculus and Matrix Applications

10 Prove by induction that $\frac{d^n}{dx^n}(xe^x) = (x+n)e^x$. What is the inductive step derivative $\frac{d}{dx}[(x+k)e^x]$?

a. $1 \cdot e^x + (x+k)e^x = (x+k+1)e^x$

b. $(x+k)e^x + e^x = (x+k)e^x$

c. $(x+1)e^x + ke^x$

d. $(x+k+1)xe^x$

11 If $A = [[1,1],[0,1]]$, prove by induction that $A^n = [[1,n],[0,1]]$. What is the matrix product $A^k \cdot A$?

a. $[[1,k],[0,1]][[1,1],[0,1]] = [[1,k+1],[0,1]]$

b. $[[1,k],[0,1]][[1,1],[0,1]] = [[1,2k],[0,1]]$

c. $[[1,k],[0,1]][[1,1],[0,1]] = [[1,k^2],[0,1]]$

d. $[[1,1],[0,1]]$

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12 A sequence is defined by $u_1 = 5$ and $u_{n+1} = 2u_n - 3$. Prove by induction that $u_n = 2^{n+1} + 3$. Assuming $u_k = 2^{k+1} + 3$, what is u_{k+1}?

a. $2(2^{k+1} + 3) - 3 = 2^{k+2} + 6 - 3 = 2^{k+2} + 3$

b. $2(2^{k+1}) - 3 = 2^{k+2} - 3$

c. $2^{k+1} + 6$

d. $2^{k+2} + 6$

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